A MULTI-LAYER DYNAMICAL ARCHITECTURE FOR HUMAN COGNITION: FROM CONTINUOUS THOUGHT TO STRUCTURAL SELF-CONSISTENCY

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ABSTRACT

This paper proposes a conception of a human cognitive architecture grounded in nonlinear dynamical ideas, using spatial-dynamical metaphors as its central framework. It constructs a three-layer structure—"Cognitive Field–Particle Flow–Microscopic Modulation Layer"—to characterize the overall relationship between thought processes, long-term biases, and individual differences. Building upon this foundation, a Gravitational Firmware Unit (GCU) is introduced as the carrier of slow variables, used to describe the long-term self-consistency of the cognitive system; a Benefit-Taxis Principle is introduced to describe a tendency toward directional selection that does not rely on explicit reward functions, but is instead guided by differences in the system's internal structural cost; a Force Field, Internal State Curves, and Benefit-Landscape structure are further introduced to characterize the mapping process from abstract personality biases to concrete behavioral trajectories. On this basis, the paper further proposes abstracting Acceleration Anchors into "the dynamical form of the Internal State," and introduces an Association Layer—carried by neural networks—as the generative mechanism of the "Directional Beacon." Within this conception, the Cognitive Field is positioned as the spatial shape presented by the overall representation of large language models; the Association Layer provides directional proposals, while the Particle Flow and Benefit-Landscape determine whether to adopt them and how to proceed; the Internal State Curve assumes the higher-order associative role of "whole-region activation," allowing the language system and the action system to share the same upstream source—language and action are thus no longer connected through instruction interfaces, but appear as simultaneous externalizations under the same Internal State. The overall tendency of this architecture is to replace the framework of "convergence optimization" with "continuous evolution"; to replace "black-box parameter fitting" with "structured conceptual variables"; to replace the intuitive image of "explicitly designed attention modules" with "focal attention emerging from the overall distribution"; to replace the decision paradigm of "numerical reward maximization" with "directional selection guided by structural cost differences"; to replace the consistency source of "context memory that can be overwritten in the short term" with "long-term biases shaped by slow variables"; and to understand "learning" and "behavior" as two aspects of the same closed-loop process. The formalization and empirical validation involved in this paper are left for future work. It must be emphasized that this paper does not claim to provide final answers to these questions, nor does it offer directly deployable engineering solutions. Rather, it seeks to preserve—alongside the mainstream wave of data-driven artificial intelligence development—a line of thinking that takes the "overall operating logic of biological intelligence" as its point of departure, and to outline the main structure of this line of thought in the clearest conceptual language possible, for reference, critique, and improvement by subsequent researchers.

Keywords *Cognitive Architecture · Dynamical Systems · Human Cognition · Multiscale Dynamics · Internal State · Self-Consistency · Decision-Making Mechanisms · Emergent Behavior*

1. Introduction

One of the central goals of cognitive science is to construct computational models capable of explaining and simulating human thought processes. Over an extended period of development, researchers have proposed a variety of cognitive modeling paradigms, including symbolism, connectionism, and various hybrid approaches [1]. These methods have demonstrated distinct advantages across different task contexts and have been widely adopted in both cognitive science and artificial intelligence.

From the perspective of modeling assumptions, different paradigms exhibit fundamental differences in their forms of representation. Symbolist approaches typically rely on explicit structures and rule-based reasoning, offering strengths in formal systems and logical consistency [2]; connectionist approaches, centered on distributed representations and statistical learning, exhibit prominent performance in pattern recognition and generalization [3]. These differences reflect modeling trade-offs under different theoretical assumptions rather than a methodological ranking of superiority.

In recent years, dynamical systems theory has gradually been introduced into cognitive modeling research, providing formal tools for describing the continuous evolution of cognitive processes [4]. Such methods treat cognitive states as dynamical evolution processes within a continuous state space, offering another perspective for characterizing temporal dependencies and context sensitivity. In neuroscience and computational modeling, attractor-based dynamical systems have been used to explain the stability of working memory and the maintenance mechanisms of neural states [5], and neural population dynamics methods have been used to characterize the continuous evolution of cortical activity.

This direction has also progressively intersected with modern machine learning methods—for example, continuous-time neural networks and Neural Ordinary Differential Equation models [6], their extensions to modeling irregular time series [7], and energy-based model frameworks [8]. Additionally, the Free Energy Principle [9] centers on the minimization of prediction error, providing a unified characterization of perception, learning, and action; the Joint Embedding Predictive Architecture [10] takes prediction in representation space as its key mechanism, offering an alternative pathway for non-generative world modeling; the predictive coding framework [11] provides a similarly intuitive neural interpretation from the perspective of cortical hierarchical structure. These advances have provided richer modeling tools while also leaving several open problems: how to introduce stable structural constraints while preserving the continuity of dynamical systems; how to coordinate short-term dynamics with long-term structure across different time scales; and how to handle cognitive aspects such as subjectivity, internal states, and personality—traditionally difficult to formalize.

Starting from these open problems, this paper proposes a conception of a human-like cognitive architecture grounded in nonlinear dynamical ideas. The purpose of this study is neither to propose a directly engineerable system, nor to establish a separate school outside of existing theories. Rather, it seeks to preserve and develop—alongside the mainstream data-driven-centered path of artificial intelligence development—a line of thinking that takes "the overall operating logic of biological intelligence" as its point of departure, and to outline its main structure at the clearest possible conceptual level for reference, critique, and further development by subsequent researchers.

2 Core Differences from Mainstream AI Models

This chapter contrasts the proposed architecture with current mainstream artificial intelligence models at the conceptual level. It should be noted that the following comparison does not constitute a rejection of mainstream models—the latter have achieved significant results in engineering deployment and task performance—but rather serves to outline the differences of this architecture in design orientation.

2.1 Objective: From Convergence Optimization to Continuous Evolution

Mainstream DNNs/LLMs are trained by minimizing a loss function, aiming to converge to a parameter state capable of stably fitting the data distribution. Once training is completed, model parameters remain largely fixed, with subsequent updates usually performed through fine-tuning.

The proposed architecture does not target a single convergence point; instead, it models the system as a continuously evolving dynamical process. During operation, the system continuously adjusts its internal structure and state distribution based on input and historical states, without a clearly defined "training completion" phase, resembling more closely a continuously updating online system.

2.2 Parameter Semantics: From Distributed Weights to Structured Variables

Mainstream models represent knowledge through high-dimensional parameters, where individual parameters typically do not carry explicit semantic meaning; their significance emerges from the statistical behavior of the entire parameter space. This approach is highly effective in engineering terms, but entails a significant cost in interpretability.

The proposed architecture divides the system state into several structured variables with explicit functional roles (such as dynamical intensity, connection density, structural position, etc.), used to describe the mechanisms underlying different aspects of cognitive processes. These variables are required to maintain relatively stable mappings to system behavior, thereby improving overall interpretability. These variables are proposed here only at the conceptual level; their concrete mathematical carriers are left for subsequent research.

2.3 Attention Mechanism: From Explicit Module to Emergent State Distribution

The attention mechanism in mainstream Transformer architectures [12] is an explicitly designed computational module that achieves selective processing of input information through weight allocation..

In the proposed architecture, information selection does not rely on an independent attention module; instead, it is jointly determined by the system's overall structural state and dynamical processes. The propagation paths and scope of influence of different inputs within the system are shaped jointly by the current structural configuration and the direction of evolution. "Being attended to" is therefore manifested as an emergent property of the system's evolution process, rather than an independent component.

2.4 Inference Method: From Forward Computation to State Evolution

The inference process of mainstream models is typically executed as forward computation under fixed parameters. Although intermediate hidden states exist during computation, they are not retained as long-term structure after inference completes.

The proposed architecture treats inference as a continuous evolution of system states. Each input not only produces an output, but also exerts influence on the current state of the system; some state changes persist into subsequent processes, enabling the accumulation and continuation of short-term states.

2.5 Reward Mechanism: From Scalar Optimization to Structural Constraints

Mainstream reinforcement learning and alignment methods rely on scalar rewards or loss functions to guide model behavior. The Free Energy Principle [9] and energy-based models [8], in different ways, take "minimization of surprise" or "minimization of energy" as their unified optimization target.

The proposed architecture does not depend on external scalar rewards, nor does it take the minimization of energy or error as its ultimate goal. Instead, it indirectly guides behavior through the system's internal structural constraints and evolutionary direction. The system tends to evolve toward directions with lower overall structural adjustment cost, where "cost" is implicitly defined by the current structural state rather than by an externally specified numerical objective. This orientation is formally similar to the Free Energy Principle (both use the descent of an internal quantity as the driving force), but differs in the object of descent—the former descends a computable quantity of surprise, whereas the latter descends structural-level reconfiguration cost, which remains a conceptual formulation in this paper.

2.6 Self-Consistency: From External Maintenance to Structural Stability

The consistency of mainstream agent systems typically depends on context windows, external memory modules, or prompt engineering, and their behavioral patterns can be significantly altered within relatively short time frames.

In the proposed architecture, the system's consistency arises primarily from internal slow-variable structures (such as core state distributions or long-term variables). These structures change little in the short term but gradually adjust over long-term operation, thereby forming stable behavioral biases and styles. This orientation bears some resemblance to the subject-level homeostasis bounded by the "Markov blanket" in the Free Energy Principle [9], but this architecture does not appeal to information-theoretic minimization formulations; rather, it emphasizes the structural sedimentation of the slow-variable layer.

2.7 Relationship Between Learning and Behavior: From Separation to Closed Loop

In mainstream models, learning (parameter updates) and inference (behavior generation) are typically treated as two relatively independent processes; the inference phase does not generally modify model parameters directly.

The proposed architecture integrates learning and behavior into a closed-loop process: behavioral outcomes continuously feed back into the system, influencing internal state distributions, enabling continuous adjustment without explicit parameter updates. Short-term changes occur mainly at the state level, while long-term changes gradually shape the core structure. This line of thought is close in intent to the "world-model continuous self-supervised learning" advocated by LeCun's autonomous machine intelligence architecture [10]; the difference is that this architecture defines "learning" as "changes in which directions within the system are regarded as beneficial," rather than as the descent of prediction error in representation space.

3 Theoretical Origin and Design Rationale of the Architecture

The primary objective of the proposed architecture is to construct a unified model capable of covering the full process of human thought and behavior. During the design process, it is necessary to balance between two extremes: on the one hand, avoiding excessive model complexity that leads to inefficient fitting of numerous microscopic variables; on the other hand, preventing oversimplification that fails to capture essential characteristics of human cognition, particularly creativity, dynamical behavior, and long-term consistency. Guided by this objective, the framework evolves progressively from initial structural hypotheses toward a dynamical modeling paradigm, ultimately forming a cognitive architecture centered on dynamical interaction processes.

3.1 Early Exploration: Limitations of the Tree-of-Thought Model

In the initial stage, the architecture design was motivated by modeling human creative cognition. A fundamental assumption is that human cognition possesses strong generative capacity: rather than being confined to a finite set of reasoning paths, it can produce highly diverse combinations and extensions within a latent space.

Based on this assumption, early attempts adopted a Tree-of-Thought structure for modeling. In this approach, the thinking process is represented as a progressively expanding tree, where nodes correspond to intermediate cognitive states and branches represent possible reasoning directions. Combined with generative models, each node can dynamically expand new content, enabling multi-path exploration in complex tasks to a certain extent

However, further analysis reveals several structural limitations of this approach: 1. Computational and memory complexity: The tree structure grows exponentially as branches expand. In practical applications, approximation strategies such as pruning, sampling, or beam search are required, which to some extent constrain the effective exploration space. 2. Limitations of the reasoning paradigm: Although node contents are dynamically generated, the overall process still operates as discrete path expansion and selection. It fundamentally belongs to a search-driven paradigm, where generative capacity is constrained by the current distribution and path expansion strategy, making it difficult to directly represent continuous state evolution. 3. Insufficient representation of dynamical properties: The Tree-of-Thought framework organizes reasoning into discrete steps, making it suitable for staged decision-making; however, it is less capable of capturing continuous transitions, cross-level jumps, and nonlinear integration of multiple factors.

Therefore, while the Tree-of-Thought approach offers clear advantages in enhancing exploration for complex tasks, its reliance on discrete path search introduces inherent limitations in modeling continuously evolving cognitive processes. This observation motivates a shift toward modeling approaches centered on state evolution..

3.2 Core Idea: From Structural Expansion to Dynamical Modeling

Building upon the analysis above, this work shifts the modeling focus from "explicit structural expansion" to "continuous evolution of states." A key observation is that human thinking does not primarily operate through the continuous explicit expansion of structure, but is better characterized as continuous state transitions and interactions within a bounded system.

Accordingly, the cognitive system is formulated as a bounded yet extensible state space. Within this space, different cognitive elements exist as dynamic states, whose evolution is jointly driven by local interactions and overall constraints. Each instance of thinking can thus be viewed as a continuous trajectory of system states within this space, rather than a path selection over predefined structures.

Within this framework, dynamical interactions—including state coupling, composition, and reconfiguration—serve as the core mechanism for generating complex cognitive behaviors. Through appropriate structural constraints and evolution rules, the system is capable of producing highly diverse behavioral patterns within a limited representational space, while maintaining overall stability and consistency. It is important to emphasize that this process is not unconstrained stochastic variation, but is regulated by a set of structural rules that determine the direction, magnitude, and scope of state changes, thereby achieving a balance between expressive capacity and controllability.

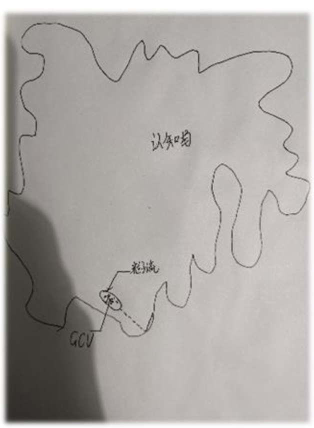
4 Core Design of the Three-Layer Dynamical Cognitive Architecture

The proposed architecture models cognition as a multi-layer dynamical system evolving in a continuous space. The overall structure consists of three coupled subsystems: the outer Cognitive Field, the intermediate Particle Flow layer, and the inner Microscopic Modulation layer (Pattern Primitives). These three layers operate at different scales: the Cognitive Field provides semantic structure and spatial constraints, the Particle Flow layer characterizes the primary trajectories of thought, and the Microscopic Modulation layer introduces fine-grained perturbations and individual variability. Together, they form a unified framework capable of describing continuous reasoning, associative jumps, and the emergence of stable behavioral patterns.

4.1 Outer Layer: Hollow-Sphere Cognitive Field

The Cognitive Field serves as the underlying space of the dynamical system. It is defined as a finite closed manifold, which can be intuitively represented as a hollow sphere, but may also be extended to higher-dimensional topological structures. Its primary role is to host cognitive and behavioral nodes while defining the boundary and semantic topology of the system.

Its key properties are as follows.



1. *Semantic definition of nodes:* Each point on the Cognitive Field corresponds to an individual cognitive or behavioral node, representing a specific concept, action, perception, or abstract thought unit.

2. *Spatial encoding of relevance:* The semantic correlation between two nodes is directly characterized by their spatial distance. The closer the distance, the stronger the correlation between nodes; the farther the distance, the weaker the correlation.

3. *Topological cognitive mapping:* Dense regions of nodes in the Cognitive Field correspond to highly related clusters of concepts, representing familiar and frequently accessed knowledge domains; sparse regions represent abstract, peripheral, or inadequately learned behaviors and concepts.

4. *Finite but extensible boundary constraint:* As a finite closed manifold, the Cognitive Field prevents the unbounded divergence of thought in unstructured space; meanwhile, as the system operates, node density gradually increases and the internal topology becomes progressively refined. The manifold itself can also be extended through projections of higher-dimensional deformations onto the observation layer.

The characterization of the Cognitive Field in this section is temporarily treated as an abstract structural conception. Its concrete correspondence with existing artificial intelligence methods—particularly the possible connection with the overall representation of large language models [12]—will be developed in Chapter 8

4.2 Intermediate Layer: Particle Flow as the Dynamical Core of Thought

The Particle Flow constitutes the dynamical core of the entire cognitive system, located within the interior of the

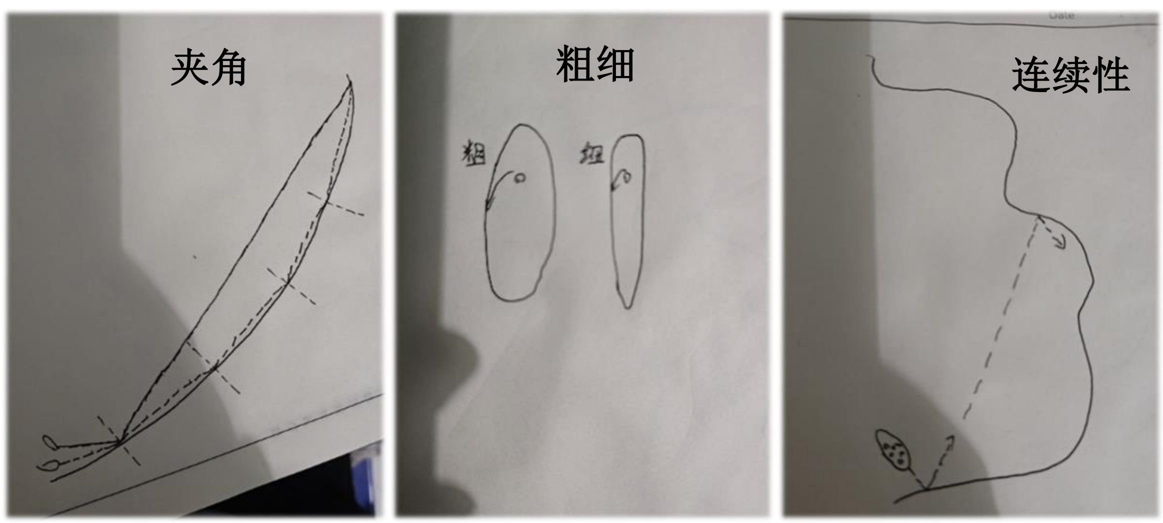
Cognitive Field hollow sphere, and is used to describe the mode of operation of human thought rather than its specific content. In the proposed architecture, the process of human thinking is modeled as a continuous motion of particle flow within the Cognitive Field, characterized by direction, inertia, and energy consumption. Its central idea is to unify logical reasoning and creative thinking through the dynamical attributes of particle flow. The Particle Flow has three core attributes:

1. *Direction and angular transitions:* The motion of the particle flow is characterized by direction, and its collision with nodes of the Cognitive Field corresponds to a completed thought event; the impacted node defines the semantic outcome of the thought. The angle between two consecutive collisions corresponds to the coherence and jumpiness of thinking—smaller angles correspond to coherent, logically consistent thought; larger angles correspond to abstract, jumpy thought.

2. *Flow thickness (inertia):* The cross-sectional magnitude of the particle flow corresponds to cognitive inertia—thick particle flows exhibit strong directional persistence, requiring higher structural cost to change direction, corresponding to focused, stable, and less easily disturbed states of thought; thin particle flows are flexible and easily redirected, with relatively lower stability and continuity.

3. *Continuity constraint:* The motion of the particle flow obeys a continuity constraint—in the absence of external input, inner-layer perturbation, or structural changes in the Cognitive Field, its direction does not change arbitrarily but evolves smoothly within a constrained range.

Through the above three attributes, the architecture provides a unified parameter space for logical thinking and creative thinking: the two are no longer viewed as opposing categories, but as different state manifestations of the same dynamical system under different parameters.



4.3 Inner Layer: Pattern Primitives as the Microscopic Modulation Layer

The Pattern Primitives are a large number of microscopic units distributed within the motion space of the particle flow. They are the smallest composable substrate of behavior and thought patterns in the cognitive system, and also a transitional layer between overt thought and underlying latent structures. Pattern Primitives themselves do not directly correspond to specific concepts or semantics; their primary function is to finely modulate thought trajectories through microscopic interactions with the particle flow, serving as the microscopic carrier of behavioral habits and thinking styles. Their key roles and properties are as follows:.

1. *Trajectory modulation:* Collisions between Pattern Primitives and particle flow alter the motion trajectory of the particle flow. A single collision produces only a minor directional offset, but the accumulation of many small offsets eventually changes the direction of macroscopic thought.

2. *Determination of cognitive characteristics:* The number and spatial distribution of Pattern Primitives determine the system's degree of caution and variability—the denser the distribution, the more frequently the particle flow is corrected, corresponding to a more cautious and thorough style of thought; the sparser the distribution, the more decisive but also more one-sided the style of thought.

3. *Exploration and stochastic perturbation:* This layer also introduces low-intensity uncertainty, enabling the system to deviate from existing paths and explore new state regions. This mechanism is an important source of novel associations and creative outcomes.

4. *Source of creativity:* The irregular, quasi-thermal motion of the Pattern Primitives introduces random microscopic perturbations into the motion of the particle flow. These perturbations are an important source of the system's creativity; small trajectory offsets may cause the particle flow to collide with Cognitive Field nodes that would otherwise be unreachable, forming breakthrough creative thinking—this intuition corresponds to the "flash of insight" in human creative thought.

The presence of the Pattern Primitive layer ensures that changes in the cognitive system do not rely solely on adjustments of macroscopic rules, thereby providing the carrier for subtle differences among individuals in thinking habits and behavioral styles—constituting the microscopic basis of individual differentiation.

5 Slow-Variable Structure and the Dynamical Mechanism of Self-Consistency

The three-layer dynamical structure is capable of characterizing continuous reasoning and jump-like associations, but relying solely on local interactions among the three layers still faces a key problem: under persistent perturbation and exploration dynamics, the system as a whole may gradually lose its stable direction, manifested as phenomena such as trajectory dispersion and inconsistent behavioral patterns. This problem corresponds to the risk of "disordered diffusion" in cognitive systems: when local dynamics lack long-term constraints, the system possesses strong exploratory capability but has difficulty maintaining consistency across time scales. To address this issue, the proposed architecture introduces a slow-variable structure that constrains the long-term evolutionary direction of the system without interfering with local dynamics, thereby achieving the unification of stability and flexibility.

5.1 Necessity of Introducing Slow Variables

Empirical observations of human cognition reveal a fundamental feature: core behavioral tendencies, value preferences, and thinking styles do not change significantly due to single inputs or short-term experiences, but exhibit strong temporal stability; in contrast, specific thoughts and behavioral responses have a much higher frequency of change. This distinction implies that cognitive systems inherently operate across multiple time scales: fast variables correspond to immediate processes of thinking, decision-making, and action; slow variables correspond to long-term structural biases (such as behavioral tendencies and value orientations). The coupling of the two enables the system to adapt flexibly in the short term while preserving consistency in the long term. Therefore, the introduction of slow-variable structures is a necessary condition for constructing stable cognitive systems.

5.2 Gravitational Firmware Unit (GCU) and Its Closed-Loop Evolution Mechanism

Based on the necessity of slow variables, this architecture introduces the Gravitational Firmware Unit (GCU) as the core long-term structural component of the system. The GCU does not directly participate in specific reasoning or decision-making; instead, it acts as a slow-variable structure that continuously influences the system's overall evolutionary direction, thereby maintaining long-term consistency without interfering with local dynamics.

5.2.1 Definition and Basic Properties of the GCU

1. *No intrinsic driving force:* The GCU does not possess direct driving capability—it does not generate specific thought trajectories or behavioral outcomes, but influences the distribution of system behavior by providing long-term biases, thereby exerting long-term constraint over the dynamics of thought. Its initial state is a neutral, unbiased state, and its eventual configuration is shaped by the system's historical experience and thought trajectories.

2. *Slow dynamics:* The evolution of the GCU exhibits time-scale separation. Compared with the fast changes of the Particle Flow layer and the Microscopic Modulation layer, the GCU adjusts only gradually under the accumulation of long-term experience, thereby forming stable structural characteristics.

3. *Indirect influence mechanism:* The mechanism of action of the GCU is indirect—it does not directly intervene in thought trajectories, but reshapes the internal structural distribution of the system, altering the accessibility and evolutionary cost of different paths.

5.2.2 Mechanism of Action and Closed-Loop Evolution of the GCU

Based on the above properties, the GCU and the three-layer dynamical structure together form a cross-time-scale closed-loop evolution mechanism that enables the system to maintain overall stability through persistent change. This mechanism can be summarized as the following process:

1. *Structural bias modulates Pattern Primitive distribution:* The GCU does not directly exert a physical "attractive force" on the Pattern Primitives; instead, it performs long-term modulation of the probabilistic weighting of spatial structure, causing the stochastic motion of the Pattern Primitives to exhibit, in the statistical sense, a biased concentration toward the structural neighborhood where the GCU resides. Accordingly, the distribution of Pattern Primitives is transformed from a uniform stochastic process into a non-uniform structural-weighted process shaped by the GCU.

2. *Distribution structure determines particle flow path cost:* The spatial distribution of the Pattern Primitives defines the "collision environment" for the particle flow, thereby determining the statistical feasibility of trajectories: high-density regions correspond to frequent interactions, supporting the continuous correction and extension of trajectories, forming stable paths with low structural cost; low-density regions lack corrective sources, resulting in less stable trajectories with higher structural cost. The motion of the particle flow is therefore not dominated by a single driving direction, but determined by the set of feasible paths constrained by the distribution structure.

3. *Behavioral distribution reshapes the GCU structure:* The long-term high-frequency trajectories of the particle flow exert a statistical reshaping on the distribution of Pattern Primitives, and this distributional change further, through a structural feedback mechanism, slowly alters the effective position of the GCU. Specifically: high-frequency thought paths → the distribution of Pattern Primitives in corresponding regions is repeatedly reinforced; reinforced distribution → alters the structural weight center; drifting weight center → gradual adjustment of the GCU's effective position. Hence, the system forms a closed-loop mechanism: behavioral trajectories → distribution structure → GCU bias → behavioral trajectories.

Through this process, the system establishes a stable cross-scale feedback: short-term behavior is determined by current state dynamics, long-term structure is shaped by behavioral distributions, and long-term structure in turn constrains future behavior. This mechanism achieves coordination across time scales, maintaining global consistency while preserving local flexibility. This closed loop is functionally similar to the self-organizing process described by the "perception–action cycle" in the Free Energy Principle [9], but this architecture does not take probabilistic inference as the mathematical basis of the closed loop; rather, it uses structural cost and distributional reshaping as its intuitive mechanism, with formalization left to subsequent research.

5.3 Unified Dynamical Origin of Self-Consistency and Creativity

Within this framework, "self-consistency" is not directly defined by explicit rules or fixed parameters, but is the emergent result of structural biases formed through long-term system evolution. This consistency does not rely on static constraints; rather, it arises from dynamically stable structures under multi-time-scale coupling.

Its main manifestations are as follows: first, under different input conditions, the system tends to select similar particle flow directions, thereby producing behavioral patterns that share overall style while retaining diversity; second, long-term structural change is gradual and does not mutate under single inputs or short-term perturbations, ensuring the stability of evolution; third, the hierarchical coupling of fast and slow variables only constrains the long-term direction of evolution, without interfering with specific thought processes and local perturbations.

Within the same mechanism framework, creativity is not suppressed by structural constraints, but is redefined. Local dynamics remain open, allowing short-term perturbations and trajectory deviations (such as stochastic perturbations of the Pattern Primitives or large-angle jumps of the particle flow), so the system retains the capacity to explore new state regions. In this framework, creativity is defined as: **recombination of states and reconstruction of trajectories under existing structural constraints, rather than unconstrained random generation**. This definition endows creativity with both openness and directionality, preventing it from degenerating into unstructured random diffusion.

Therefore, in this architecture, long-term structural bias does not play the role of a "limitation," but rather of a boundary-regulating structure—providing a stable reference frame for exploration so that generated outcomes retain diversity while possessing interpretable directionality and utility value, thereby unifying stability and creativity within the same framework.

Summary: By introducing the GCU as a slow-variable core, the proposed architecture establishes a cross-time-scale regulatory mechanism within a continuous dynamical system. This mechanism allows the system to maintain short-term dynamical flexibility while forming long-term stable behavioral biases, thereby achieving self-consistency and sustainable evolution in cognitive processes.

6 Benefit-Taxis Principle and Directional Decision-Making Mechanism

The three-layer dynamical structure and the GCU mechanism together construct the state space and the long-term consistency mechanism of the cognitive system; however, dynamical evolution alone still cannot explain a key problem: when the system resides in a state where multiple reachable paths coexist, what is the origin of its selection mechanism? In other words, the dynamical structure describes "what can happen," but cannot by itself explain "why a particular path is selected." Therefore, this architecture further introduces the Benefit-Taxis Principle, which provides the system with a rule for directional selection, transforming it from a neutral dynamical system into a cognitive system with an intrinsic preference structure.

6.1 Basic Definition of the Benefit-Taxis Principle

The Benefit-Taxis Principle is not an external reward function or an explicit utility-maximization mechanism, but a directional selection rule determined by the system's internal structure. Its core definition is: the cognitive system tends to continue evolving along the direction that is most consistent with its current dynamical state and that incurs the lowest structural evolution cost. Under this definition, "decision" no longer refers to the selection of discrete actions, but to the determination of the next direction of evolution within the state space. This principle entails three basic reformulations: "benefit" is no longer an externally assigned scalar, but a property of coherence implicitly defined by the system's internal structure; decision-making does not correspond to a single output action, but to the choice of direction of state evolution; the Benefit-Taxis Principle operates at the level of the dynamical system, influencing trajectory evolution paths through differences in structural cost.

Unlike the mechanism of mainstream reinforcement learning [13] based on scalar rewards, the Benefit-Taxis Principle does not provide a directly optimizable objective; unlike the mathematical formulation of "minimizing free energy" in the Free Energy Principle [9], the Benefit-Taxis Principle does not appeal to probabilistic inference or energy functionals, but operates at the intuitive level of structural cost and distributional reshaping. The cost of this is the loss of optimizability; the potential benefit is the ability to offer a conceptual treatment of the long-standing question of "intrinsic motivation" from the perspective of the system's internal structure**.**

6.2 Structural Origins of Benefit and the Direction-Selection Mechanism

In this architecture, benefit is not determined by any single variable, but arises from the joint action of the system's internal multi-layer structures. Its essence is expressed as a tendency toward the overall minimization of structural cost across different temporal and spatial scales. Specifically, this structural origin consists of three types of mechanisms that jointly determine the system's direction-selection behavior.

1. *Inertial direction of particle flow:* The dynamical inertial structure determines the system's short-term evolutionary preference. The current direction of motion of the particle flow represents the system's immediate inertial state; continuation along this direction typically requires only minimal structural adjustment and therefore has the lowest local cost, constituting the system's most fundamental source of continuity..

2. *Implicit bias of the GCU:* The long-term bias structure formed by the GCU provides cross-time-scale directional constraint. As a slow-variable core, the structural state represented by the GCU determines the stable tendency of long-term evolution. Paths consistent with this bias have lower cost in terms of overall structural adjustment, whereas paths deviating from this bias require a higher degree of system reconfiguration and therefore have a significant selection advantage over long time scales.

3. *Local perturbation cost of the Pattern Primitive layer:* The local perturbation structure determines the accessibility and smoothness of a path at the microscopic level. The Microscopic Modulation layer (such as the Pattern Primitive distribution) forms local structural density; its spatial distribution influences the local perturbation cost in different directions. Regions with more continuous and denser structural distribution correspond to lower path perturbation cost; conversely, lower density implies higher uncertainty and greater adjustment cost.

4. *Directional Beacon proposal from the Association Layer:* When the particle flow makes contact with a given

node, the Association Layer carried by the Cognitive Field (provided by neural networks) presents an adjacent region as possible targets, continuously offering proposals for direction selection. The first three categories characterize "how the trajectory evolves within the current scope," while this fourth category characterizes "where it is worth going"; the Directional Beacon only offers a proposal, whereas whether to adopt it remains jointly adjudicated by the structural constraints of the first three categories. A full development of the Association Layer mechanism appears in Chapter 8.

In summary, benefit can be understood as the superposition of the above four categories of structural constraints under the current state—the system naturally tends to select the direction of evolution with the lowest overall structural cost. From a dynamical perspective, direction selection can be reduced to a trade-off between "magnitude of turn" and "structural cost": small-magnitude turns correspond to low-cost local continuous evolution and are the default mode; large-magnitude turns imply significant structural reconfiguration and usually occur only when long-term gain significantly outweighs short-term cost. This mechanism provides a consistent explanatory framework for both continuous thought and jump-like thought, and can also be used to intuitively understand phenomena such as "difficulty in resuming thought after it has been interrupted" and "efficiency differences between familiar and unfamiliar tasks.".

6.3 Temporal Structure of Benefit and Continuous Learning Mechanism

The Benefit-Taxis Principle in this architecture exhibits significant temporal hierarchical features, whose core can be unified into two mutually coupled decision levels: local benefit and global benefit. These two levels are not independent rules, but responses to structural cost at different temporal scales within the same dynamical framework.

Local benefit corresponds to the mechanism of immediate selection on short time scales, whose main feature is prioritizing evolutionary paths with low structural cost and low perturbation, thereby ensuring the system's capacity for stable response in rapidly changing environments. This level dominates most of the system's high-frequency decision processes and determines the continuity and immediate adaptability of the cognitive system under local states.

By contrast, global benefit corresponds to the structural-optimization mechanism on long time scales, whose concern is the stability and consistency of the direction of the system's overall evolution. At this level, the system allows for the short-term selection of higher-structural-cost paths in exchange for improvement of long-term structural states and optimization of overall biases. The main carrier of this mechanism is the GCU, which achieves direction stability across time scales by slowly adjusting the system's long-term structural bias.

Local benefit and global benefit together constitute a unified decision framework, enabling the system to form a dynamical balance between short-term adaptability and long-term consistency, thereby avoiding behavioral degradation under a single time scale.

Above this hierarchy, the Benefit-Taxis Principle reshapes the notion of learning. Learning is no longer defined as error correction or loss minimization, but as: the process by which certain directions within the system come to be regarded as beneficial. Every act of direction selection alters the inertia of the particle flow, the distribution of Pattern Primitives, and the cost distribution of subsequent paths, thereby forming temporal path dependence and cumulative effects. Repeated behavioral patterns gradually lower the structural cost of the corresponding paths, making other paths relatively more "expensive," and thus statistically exhibiting stable style and directional consistency. This process can be summarized in a single closed-loop relation: behavior shapes structure, and structure guides behavior.

7 Dynamical Mapping Mechanism Between Personality Dimensions and Behavioral Patterns

In this architecture, the GCU, as a slow-variable structure, determines the long-term evolutionary direction of the system, while the Particle Flow and the Microscopic Modulation layer are responsible for the specific generation of thought and behavior. However, relying solely on these two types of structures is still insufficient to explain a key question: how are abstract personality biases translated into concrete behavioral trajectories?

To this end, this chapter introduces the Force Field, the Acceleration Anchor mechanism, and the Benefit-Landscape structure to construct a continuous dynamical mapping from personality to behavior, enabling abstract structural biases to be transformed into evolvable behavioral trajectories within the state space.

7.1 Force Field: Spatial Manifestation of Personality Structure

To provide an operational expression of personality bias, this architecture introduces the Force Field as an intermediate representational structure. The Force Field is jointly determined by the spatial position of the GCU, the distributional state of the Microscopic Modulation layer, and the dynamical attributes of the particle flow; its essence is a structural potential field within the cognitive space.

The Force Field is not externally imposed, but is a natural manifestation of the system's internal state. Its main features are as follows:

1. *The shape of the Force Field is jointly determined by the system's core structures:* The overall form of the Force Field is jointly determined by multiple layers of structure: the GCU determines the global bias direction, the Microscopic Modulation layer determines local structural undulations, and the inertia and thickness of the particle flow determine the system's responsiveness to structural changes. As the system operates, local behaviors gradually alter the structure of the Force Field, while long-term stable behavioral patterns form stable structural depressions or elevations within the Force Field, thereby becoming consolidated as behavioral preferences.
2. *The complexity of the Force Field corresponds to the richness of personality:* The structural complexity of the Force Field reflects the expressive capacity of personality structure. Richer microscopic structure and higher dynamical degrees of freedom produce more complex Force Field forms, corresponding to finer-grained capacity for behavioral differentiation; conversely, it manifests as a simpler and more stable behavioral pattern—for example, the more Pattern Primitives there are, and the greater the dynamical range of the particle flow, the more complex the concave-convex structure the Force Field can form.
3. *The uniqueness of the Force Field corresponds to individual personality differences:* Different experiences and different thought trajectories shape entirely different Force Field configurations. Even two systems with identical initial states will, after different experiential inputs, form entirely different Force Fields.

The spatial position of the GCU alone cannot describe the complexity of personality; the Force Field, shaped jointly by structural biases, slow variables, interaction dynamics, and external feedback, provides a compact yet expressive dynamical representation of personality.

7.2 Acceleration Anchors: Intrinsic Localization Mechanism of Thought Paths

Building upon the Force Field representation, this architecture introduces the Acceleration Anchor mechanism to address the problem of accessing and reactivating thought paths. Unlike traditional approaches that rely on explicit coordinates or symbolic indexing, this mechanism does not depend on absolute spatial positions; instead, it uses patterns of acceleration change in the motion of the particle flow as path identifiers—what the system records is a dynamic pattern of change rather than a static node position.

The core idea of this mechanism is that path recognition does not depend on exact matching, but on similarity of dynamical features. When the system again experiences a similar pattern of acceleration change, the corresponding path can be reactivated, achieving rapid recall of thought. In colloquial terms, what the system "remembers" is the "feel" of a thought process; when a similar feel recurs, the corresponding thought path is naturally brought back.

This mechanism endows the system with two key capabilities: first, compressed expression of experiential memory, i.e., replacing explicit storage through dynamical features; second, recoverability of paths, i.e., even when the structure has temporarily weakened, the corresponding behavioral path can still be reactivated through dynamical anchors.

The positioning of Acceleration Anchors in this section focuses on path recognition and reactivation. In Chapter 8, this acceleration feature will be further abstracted into the "dynamical form of the Internal State," taking on a more general role—serving not only as a basis for path recognition, but also as a shared upstream element running through both the language channel and the action channel.

7.3 Benefit-Landscape Structure: Spatial Representation of Decision Paths

On the basis of the Force Field structure, after introducing the Benefit-Taxis Principle, the structural cost of the system can be further mapped into a scalar field within the cognitive space, forming the Benefit-Landscape structure. Within this structure, different directions correspond to different levels of structural cost, and the evolution process of the system can be understood as a path selection process within this landscape field. This structure possesses the following key properties:

1. *Landscape height is determined by structural cost:* Low-cost regions correspond to low elevation, and high-cost regions correspond to high elevation. The evolution process of the particle flow is manifested as descent motion within this landscape structure, i.e., prioritizing low-cost paths.

2. *Path selection exhibits reinforcement effects:* The local structural cost of frequently used paths gradually decreases, forming path-dependent structures; paths that have long been unused, since the plastic space of the Force Field is finite, gradually increase in structural cost, manifested as natural weakening and forgetting. However, because of the presence of Acceleration Anchors, such paths can still rapidly recover after reactivation.

3. *Intersection phenomena of neural associations:* When multiple low-cost paths intersect, they form structural junctions. These junctions correspond to mechanisms of automatic transition among different cognitive paths, and are the structural basis of association and cross-domain thinking.

4. *Landscape reshaping via longitudinal benefit:* The goal of longitudinal benefit is, through short-term high-cost behavior (i.e., actively advancing toward high-landscape paths), to reshape the long-term form of the entire landscape map, eventually forming a path structure that better fits long-term global benefit.

7.4 Unified Mechanism of Personality–Behavior Mapping

In summary, the GCU provides long-term structural bias, the Force Field provides spatial expression, the Acceleration Anchors provide a path memory mechanism, and the Benefit-Landscape structure provides decision rules. Together, the four constitute a complete mapping chain:

Personality (GCU) → Spatial Field → Dynamic Trajectory → Landscape-Based Decision → Behavioral Output

Within this framework, behavior is no longer an independently generated outcome, but the result of path evolution under structural constraints; personality is likewise no longer an abstract description, but a stable bias structure within a dynamical system.

Thus, the processes of thinking, decision-making, and behavior within the system are unified as continuous dynamical evolution within a structured landscape field.

8 Internal State, Directional Beacon, and the Kernel Unification of the Language–Action System

Chapter 7 established the mapping chain from personality structure to behavioral output, but an implicit problem remains unaddressed: for a complete cognitive subject, its language system and action system differ significantly in the type of information, in dimensionality, and in temporal granularity, yet the subject is able to naturally integrate the two under a single personality. This chapter further abstracts the Acceleration Anchor from the "path-recognition mechanism" of Chapter 7 into the "dynamical form of the Internal State," and introduces the Association Layer—carried by neural networks—as the generative mechanism of the "Directional Beacon," in order to characterize the integration of language and action within the same subject, and to clarify the design orientation of this architecture in contrast to existing embodied-intelligence paradigms.

8.1 The Translation Problem Between Language System and Action System

For any intelligent subject, the language system and the action system differ fundamentally in input/output form: language operates on discrete symbol sequences with highly compressed semantic structure; action operates on continuous physical quantities (displacement, velocity, torque, etc.) with highly distributed perception–motor coupling. Nevertheless, in human subjects, these two formally heterogeneous systems can be naturally integrated by the same individual, manifested as the coherent behavior and expressive style of "the same person.".

If the two were directly interfaced—for example, by letting language output directly drive action execution—then information transmission would inevitably degenerate into a unidirectional linear chain of instructions, presenting a mechanical response of "completing specified actions according to specified steps," unable to express the difference and totality that a subject should exhibit in different situations. It can thus be inferred that, between the two heterogeneous systems, there must exist a mediating carrier capable of performing translation and integration at the kernel level; this carrier cannot be another symbolic system, for otherwise the problem would merely be recursed one level—true integration must occur at a non-symbolic structural level.

Furthermore, the volume of information faced by this mediating carrier far exceeds what it can directly process. Taking pain as an example: if the system responded one-to-one to the microscopic variation of every nerve ending, then the event of "being struck" would be unfolded into a vast sequence of signal transmissions, rather than a "pain" that can be integrally grasped by the subject. This indicates that the mediating carrier not only performs translation but must also perform information filtering and integration—compressing high-dimensional, high-frequency, distributed low-level information into low-dimensional, low-frequency, structural features that can be integrally grasped by the subject. In this architecture, this filtered and integrated structural feature is named the "Internal State".

8.2 Neural Network Directional Beacon: The Association Layer of the Cognitive Field

Before unfolding the concept of Internal State, it is necessary first to locate the mechanism that, in this architecture, is responsible for the "association between things." In this architecture, the work of connecting things to things is primarily carried out by neural networks—including contemporary large language models. The Cognitive Field described in Chapter 4 here receives a more concrete correspondence: it is positioned as the spatial shape presented by the overall representation of large language models. Semantic distance between nodes, clustering distribution between concepts, and learned associative structure can all be carried at this layer. Therefore, this architecture does not attempt to establish another layer of associative structure outside of neural networks, but regards existing neural networks as the natural implementation form of the Association Layer.

However, the Association Layer alone cannot support a cognitive system with subjectivity. The reason is that the direction and angle of the particle flow are jointly determined by the Force Field and the Benefit-Landscape; both of these are local judgments about "how the trajectory evolves within the current scope," and do not answer the question of "where the particle flow should go and how far." The neural association of the Cognitive Field plays a role precisely at this position: when the particle flow makes contact with a given node, the associative structure carried by the neural network presents an adjacent region as the "next possible target," forming continuous directional guidance for the particle flow. This architecture names the directional guidance provided by the Association Layer the Directional Beacon.

The Directional Beacon serves only a "proposal" role—indicating which directions are possible in the associative sense, without determining whether the system adopts them at the moment. Whether to go, how to go, and whether to jump directly or first take a detour, remain jointly determined by the characteristics of the particle flow and the Benefit-Landscape. A concrete example: while reading, one happens to see a certain figure; this figure triggers a related region in the Association Layer and forms a Directional Beacon; if the landscape indicates that this region is of high cost (unfamiliar, or highly disruptive to the current reading), lateral benefit pulls the particle flow back to the text; if the cost is low, the particle flow follows the Directional Beacon and jumps to thinking about that figure. The Beacon proposes, the particle flow disposes—this is the basic division of labor in this architecture between the Association Layer and the dynamical layer.

8.3 Limitations of the Directional Beacon and the Need for Higher-Level Association

The Directional Beacon mechanism described above handles connections between "things and things," but this level of association is still insufficient to explain another class of phenomena in everyday cognition: the triggering of a concept often simultaneously activates an entire region, rather than lighting up nodes one by one in serial order.

Take "clenching the fist" as an example. When a person hears this phrase, one does not process node-level instructions in the mind such as "bend the middle finger by how many degrees, bend the index finger by how many degrees"—these details are almost never explicitly invoked in the cognitive process. What one does, at the moment of receiving this concept, is to cause an entire region of related action nodes to be activated simultaneously and executed in concert. This phenomenon suggests that this architecture needs to introduce a higher-level association mechanism—one that connects not "nodes with nodes," but rather "some overall feature" with "an entire region." This higher-level feature cannot continue to reside in a discrete-node representation, but requires some form of continuously evolving expression possessing an overall trend.

The Internal State Curve introduced in the next section is the conceptual conjecture of this form of expression.

8.4 Internal State Curve: A High-Dimensional Integrative Form

In this architecture, the Internal State can conceptually be characterized as: a second-order variation pattern of the system's internal structural state over time, namely a curve formed by a series of acceleration features. Different motion trajectories of the particle flow, through the closed-loop mechanism, exert structural reshaping on the GCU and the Microscopic Modulation layer; such reshaping is continuous and dynamic, and therefore naturally possesses a temporal velocity characteristic; the variation of velocity constitutes acceleration, and the acceleration curve of the system state over time is conceived as a high-order integrative expression of the underlying high-dimensional information.

This definition entails three key tendencies. First, the Internal State is conceptually dimensionality-reduced—large amounts of detail in the original input are naturally filtered during upward transmission, retaining only those parts that actually perturb the structure. Second, the Internal State is not information loss, but extraction of the essence of information—what is retained is the most structurally meaningful variational feature of the original input. Third, the Internal State is conceptually modality-independent—regardless of whether the input comes from language symbols or physical sensors, as long as it induces a similar acceleration curve within the structural space, it corresponds to a similar type of Internal State. The Internal State thereby becomes a natural carrier for cross-modal integration. This orientation of "cross-modal integration of representations" is intuitively similar to JEPA [10]'s handling of heterogeneous inputs in representation space, but this architecture does not take prediction error in representation space as the training target; instead, it regards the acceleration curve as a characteristic pattern naturally arising from the structural-evolution process.

With the introduction of the Internal State, the "concept activating an entire region" problem posed in Section 8.3 can be addressed. This architecture conjectures: the Association Layer carried by neural networks, in addition to learning the association between "things and things," also learns the association between the "form of Internal State Curves" and "an entire region." Which curve shape corresponds to which region of nodes is not preset, but is gradually established through association within experience.

This learning process can be intuitively understood through two examples. First, a newborn, upon hearing "make a fist" for the first time, does not possess the corresponding action response; as an adult teaches them step by step, initially they can only awkwardly and one by one curl their fingers; but with repetition, the association between the Internal State trend triggered by the concept "make a fist" and the entire region of hand-action nodes is gradually established; thereafter, whenever an Internal State of similar trend reappears, the Directional Beacon points to that entire region of nodes, and the particle flow is guided across the region as a whole—the reflexive completion upon again hearing "make a fist" is the natural outcome of this association having matured. Second, when a person hears the concept "family," what is activated is not a single node, but a trend of Internal State of a particular form; different sub-features of this trend are respectively associated with specific family members; the particle flow travels along the direction indicated by each sub-feature, and each direction corresponds to a more specific person and scene. A single word activates an entire region, but the mode of activation is not list-style enumeration; it is rather a one-time presentation through the high-dimensional feature of the Internal State.

The characterization of the "Internal State" in this paper involves only the functional level—used to describe the way it acts within the structure of the system—and does not involve the question of whether or how subjective experience exists or arises. The latter belongs to the domain of consciousness research and lies beyond the discussion scope of this architecture.

8.5 An Intuitive Typology of Internal State Curves

Building on the above, different forms of the acceleration curve can correspond to different types of Internal State characteristics. This section offers several intuitive types as a preliminary illustration of the spectrum of Internal States, not as a complete classification.

1. ***Smooth-type curves:*** When the external input is consistent with the current direction of the particle flow and the long-term bias of the GCU, the system, under the Benefit-Taxis Principle, requires no large-scale structural adjustment, and the acceleration curve tends to be smooth and stable. This form corresponds to Internal States such as calm, focus, and flow—its features are low rate of structural change and low variation of the rate itself
2. ***Abrupt-type curves:*** When the external input produces strong perturbations not easily accommodated by the existing structure (such as pain, shock, or strongly conflicting information), the Microscopic Modulation layer must undergo large-scale reconfiguration to restore structural self-consistency, and the acceleration curve exhibits

significant abrupt changes over time. This form corresponds to high-intensity Internal States such as pain, shock, and anger.

3. ***Oscillatory-type curves:*** When the system repeatedly switches between two competing structural biases, the acceleration curve exhibits periodic undulations, corresponding to Internal States such as entanglement, hesitation, and contradiction—whose essence is the particle flow's repeated probing among several nearby low-valleys within the Benefit-Landscape.

4. ***Slow-rising-type curves:*** When the system is long subjected to persistent input inconsistent with the GCU bias, the acceleration curve slowly but continuously deviates from the zero level, corresponding to Internal States such as fatigue, stress accumulation, and latent discomfort—whose features are that a single perturbation is not significant, but accumulated effects are significant.

5. ***Suddenly-flat-type curves:*** After prolonged oscillation or rise, the system suddenly enters a stable state, and the acceleration curve shifts from high amplitude to near flatness, corresponding to Internal States such as epiphany, relief, and decision made—whose essence is the system's transition from multiple metastable structures to a lower-cost stable configuration.

The core claim of this architecture is: any Internal State that can be integrally grasped by the subject can in principle find a corresponding form within the space of acceleration curves; differences in Internal States among individuals manifest as differences in the cost distribution shaped by their GCU and Force Field. It should be particularly noted that the five types above serve only as intuitive entry points and should not be understood as a complete classification—the acceleration curves produced by a real system within any short interval have continuously infinite forms, and almost never recur exactly. Therefore, this architecture does not adopt discretized formulations such as "action family" or "feeling family," in order to avoid the misimpression of a pre-enumerated correspondence table.

8.6 An Intuitive Typology of Internal State Curves

On the basis of the above associations, the response processes between Internal States and language/action can be further described. This process is conceived as bidirectional in this architecture—an Internal State driven by external stimulus can trigger the corresponding language or action output, and the representation carried by a language input can also trigger the corresponding Internal State response. Two examples illustrate this bidirectionality.

Direction 1: From external stimulus to language output. When the subject is touched by an external stimulus—for example, being pinched—the system produces a corresponding Internal State trend; the Association Layer of the neural network associates this trend with the word representation "pain"; under the joint action of the GCU bias and the landscape structure, the particle flow is guided toward the region related to articulation; at this point, the Internal State trend is further associated with specific articulatory control patterns, and is ultimately externalized in linguistic form as "pain." The entire process does not rely on any explicit instruction—every stage is jointly advanced by the Directional Beacon of the Association Layer and the dynamics of the particle flow.

Direction 2: From language input to concretization of the Internal State. When the subject hears someone else say the word "pain," the representation of this word first triggers the Association Layer, producing a Directional Beacon; according to the current landscape, the particle flow may be guided in that direction. If this Directional Beacon has a deeper association with some Internal State in the subject's experience, then, along the way there, the subject produces an Internal State trend of similar form (often only similar, and not necessarily of high intensity); if the association is deep enough—for example, the subject has recently experienced a similar pain, or the speaker is closely related to the subject—then the Internal State is actually activated, and in turn forms a response with specific nodes in the Association Layer. At this point, what the Directional Beacon points to is no longer the representation of the word "pain" itself, but is concretized into specific events, specific scenes, or specific persons within the subject's experience.

The symmetry of the two directions suggests: the relationship between Internal State and specific nodes is not a one-way mapping, but a bidirectionally activated and mutually shaping coupling. What the neural network learns is not a single direction of "predicting nodes from Internal State" or "predicting Internal State from nodes," but a continually adjusted bidirectional association between the two. The deeper the coupling, the more closely the responses correspond in both directions.

8.7 Returning to the Embodied Interface: From Instruction Translation to Kernel Translation

Combining the mechanisms above, one can answer the interface problem posed at the beginning of this chapter—in what form the connection between language system and robot should be carried out.

In the traditional paradigm of embodied intelligence, the interface is carried by instructions: a language model generates a structured string of instructions, and the controller executes them step by step. Under conditions of clear tasks and controlled scenes, this paradigm offers good engineering reliability. However, its inherent limitation at the design level lies in the fact that information transmission between high and low levels is unidirectional, symbolic, and "de-Internal-State"; the system lacks an internal subject running through both language and action, and it is therefore difficult to manifest individual differences and context-dependent integral responses at the subject level.

In this architecture, the interface is no longer carried by instructions, but by the "association between Internal State Curves and an entire region of neural action nodes": the Internal State Curve serves as the target Directional Beacon for the GCU-Pattern Primitive system; the language system generates expressions compatible with the current Internal State; the action system activates the entire region of action nodes associated with that same Internal State. The coordination of the two is not achieved by instruction stitching, but by sharing the same upstream—the Directional Beacon is the common source of both, and the Directional Beacon is generated by the Association Layer jointly from the Internal State and external signals.

The direct consequence is: the robot's real-time behavior is no longer described as "raise the arm by 30 degrees, rotate by 45 degrees, shift the center of gravity to a certain point," but is described as—"just woke up, stretching, quite angry, bursting into scolding." Each such phrase corresponds to an Internal State trend, corresponds to an entire region of action nodes activated simultaneously, and corresponds to a situation simultaneously expressed by the language system. Under this design, there is no "translation" step between language and action, because the two are not in a sequential relationship but are simultaneous externalizations of the same Internal State in two output channels.

This architecture simultaneously regards language and action as externalizations of the GCU kernel on different output channels: the language model bears the computation and linguistic expression of the semantic layer; the physical interface bears the information input and behavior execution of the perception–action layer; the GCU and its associated acceleration curve serve as the kernel carrier running through both, responsible for the formation, integration, and directional guidance of the Internal State; the neural-network Association Layer then runs through the whole as the generative mechanism of the Directional Beacon. The relation among the four is no longer that of invocation between modules, but the unified evolution of the same dynamical system at different scales. This conception corresponds to the intuitive image of a complete living activity: the physical layer corresponds to the senses, the language–semantic layer corresponds to cognition, the GCU kernel corresponds to the self and the Internal State, and the Association Layer corresponds to the mechanism by which experiential connections are consolidated.

9 Discussion, Limitations, and Conclusion

This study proposes a human-like cognitive modeling framework based on nonlinear dynamics, organized around the core of "Cognitive Field–Particle Flow–Microscopic Modulation structure," introducing a slow-variable structure (GCU) and the Benefit-Taxis Principle to describe behavioral differences and directional tendencies of cognitive systems across different time scales, and using the Acceleration Anchor and Internal State filtering mechanism to outline the integration of the two heterogeneous channels of language and action within a single subject. On this basis, the Cognitive Field is positioned as the spatial shape presented by the overall representation of large language models, and the Association Layer carried by neural networks is taken as the generative mechanism of the "Directional Beacon," so that the architecture is connected at specific points with contemporary neural-network methods. Overall, this architecture attempts to provide, from a dynamical perspective, a unified structured account of the relations among thought, Internal State, learning, and behavior.

9.1 Key Innovations and Methodological Contributions

The main contributions of this research can be summarized as follows:

First, this architecture unifies cognitive processes as a continuously evolving dynamical picture, rather than as traditional discrete symbol processing or purely probabilistic modeling. Under this perspective, thinking, reasoning, and decision-making are narrated as manifestations of the same system across different time scales, thereby providing an integrative conceptual framework.

Second, this architecture introduces a hierarchical coupling narrative between the slow-variable structure (GCU) and fast-variable processes, used to characterize the relation between long-term stability and short-term flexibility, providing an account of "how long-term preferences influence short-term behavior" that does not depend on external memory modules or explicit rules.

Third, at the decision-making level, this architecture proposes the Benefit-Taxis Principle, narrating "decision" as "directional selection guided by structural-cost differences"—taking "internal structural coherence" in place of an "external value function" as the source of preference.

Fourth, the Force Field structure and Benefit-Landscape expression provide a mediator between abstract biases and concrete behavioral trajectories—making personality preference no longer merely a static label, but something that can be understood as part of a dynamically evolving structure.

Fifth, this architecture explicitly positions the Cognitive Field as the spatial shape presented by the overall representation of large language models, and takes the Association Layer carried by neural networks as the generative mechanism of the "Directional Beacon." The Directional Beacon bears only the proposal of direction, while "whether to go and how to go" remains jointly adjudicated by the particle flow and the Benefit-Landscape. This division of labor provides a specific path for connecting this architecture with contemporary neural-network methods (such as the Transformer architecture [12], active inference in the Free Energy Principle [9], and JEPA [10])—without establishing another layer of associative structure, but placing existing methods in the associative position within the dynamical system.

Sixth, beyond the Association Layer, this architecture further introduces the acceleration curve as the conceptual form of the "Internal State," conceptualizing the Internal State as a second-order variation pattern of the system's structural state over time. What this expression bears is not only the provision of a shared "Directional Beacon" for language and action, but also a candidate response to the question of "why a concept is able to activate an entire region"—the neural-network Association Layer learns not only the correspondence between things and things, but also, through experience, gradually establishes the association between the form of the Internal State Curve and an entire region of nodes. Thereby, within this architecture, the language system and the action system are no longer connected through instructions, but appear as simultaneous externalizations under the same Internal State.

Seventh, this architecture characterizes the relation between Internal States and specific nodes as bidirectionally responsive coupling—external stimuli can induce Internal State trends and, through the Association Layer, externalize them as language or action; language inputs can also trigger Directional Beacons and, under deep association, in turn be concretized into the unfolding of the Internal State. This bidirectionality provides a candidate path at the structural level for the explanation of "subjectivity," and also contributes one line of continuable questioning to the long-standing discussion point of "why embodied intelligence lacks a sense of subjectivity"..

9.2 Current Limitations and Implementation Challenges

It should be noted that this study remains at a conceptual and theoretical stage, with several limitations in formalization and practical implementation.

First, at the mathematical level, this research still remains at the stage of conceptual narration. Expressions used in the text such as "structural cost," "slow-variable evolution," "acceleration curve," "Benefit-Landscape," "Directional Beacon," and "Association Layer" are all intuitive concepts and have not yet been assigned strict mathematical objects. Any phrasing in this paper that appears to carry formal meaning should not be regarded as a mathematical claim, nor is it sufficient to support quantitative conclusions.

Second, the multi-time-scale coupling, microscopic modulation, slow-variable evolution, and Directional Beacon generation envisioned in this architecture, once numerical implementation is considered, will face considerable computational complexity and stability challenges; no corresponding efficient implementation scheme is currently available.

Third, regarding the core hypothesis of "cross-modal structural perturbation mapping"—that is, that inputs from any modality can be mapped into structural perturbations in the GCU space and integrated at the level of the acceleration curve—its mathematical form and realizability remain to be determined. Directions such as topological data analysis, information geometry, and tangent-space projection may serve as candidates, but all remain to be verified

Fourth, this architecture corresponds the "Cognitive Field" to the spatial shape presented by the overall representation of large language models, and corresponds the Association Layer to the learning capability of neural networks. This correspondence is still only a conceptual analogy in this paper; the degree of agreement with the actual representational structure of existing neural networks, and whether it can support the "Directional Beacon–Particle Flow–Internal State" division of labor envisioned in this architecture, both lack corresponding empirical research.

Fifth, concerning the conjecture that "the association between the Internal State Curve and an entire region of nodes is established by neural networks through experiential learning," this paper provides only intuitive examples such as "an infant learning to make a fist" and "hearing 'family,'" and does not provide a verifiable learning process, training signals, or convergence metrics. The mode of engineering implementation of this mechanism, the feedback signals required, and how the Internal State itself can stably serve as input to neural networks, all await further research.

Sixth, this architecture currently lacks a standardized evaluation system. Concepts such as "personality consistency," "long-term preference stability," "Internal State plausibility," "Directional Beacon guidance efficacy," "embodied consistency," and "creativity performance" are all descriptive expressions in this paper; how to transform them into quantifiable and comparable metrics is a task that subsequent work must face.

Seventh, the characterization of the "Internal State" in this architecture involves only the functional level, not the consciousness question of whether or how subjective experience exists or arises. This question constitutes the natural terminus of this framework in explaining subjectivity; any extended interpretation that "this architecture can produce consciousness" is outside the scope of this paper's argumentation.

In summary, this paper should in no sense be understood as research that has "resolved" the above problems, but is merely an attempt, in a relatively unified conceptual language, to reorganize these problems, so that subsequent work may proceed along clearer lines.

9.3 Directions for Future Research

On the basis of the above limitations, several concrete directions for subsequent research can be outlined, offered only for reference to subsequent researchers.

First, mathematical formulation of GCU slow-variable evolution. The focus is on seeking suitable mathematical tools (such as slow–fast system theory in nonlinear dynamics, bifurcation theory, attractor theory, and renormalization group methods), providing a formal description of the long-term evolution of the GCU and characterizing its stable states (corresponding to personality stability) and state transitions (corresponding to personality formation and change) with corresponding features.

Second, concretization of the cross-modal structural perturbation mapping. Explore operator forms for compressing inputs of arbitrary modality into structural perturbations (such as topological data analysis, information geometry, and tangent-space projection), and verify the "modality-independence" intuition on the simplest feasible instances.

Third, research on the correspondence between the Cognitive Field and the representations of large language models. Examine in what sense the internal representations of existing large language models carry the node distribution, distance measure, and reachability structure required by this architecture, and attempt to introduce slow-variable mechanisms (such as low-rank adaptation weights, trainable prompts, or dedicated state channels) as carriers of the GCU, exploring modes of bidirectional feedback between the output of language models and dynamical states.

Fourth, the implementation form of the Association Layer and the Directional Beacon mechanism. Study the feasibility of a dedicated neural-network module that specifically bears "the association between the Internal State Curve and an entire region of nodes," examining its mode of cooperation with existing large models, the forms of training signals required, and methods for evaluating the stability of the associations. This direction is a key step in advancing the architecture from conceptual conception to a runnable prototype.

Fifth, embodied experiments on the Internal State Curve and "whole-region activation." On simplified robotic or simulated platforms, attempt to realize the hypothesis that "the same Internal State Curve corresponds to multiple specific actions," and observe whether, under such a mechanism, language expression and action expression exhibit consistency tendencies at the subject level, and whether the bidirectional response mechanism (external stimulus → Internal State → language/action; language input → Directional Beacon → concretized Internal State) can be observed in the simplest cases.

Sixth, establishment of an evaluation system. Develop quantifiable metrics around dimensions such as "personality consistency," "Internal State plausibility," "Directional Beacon guidance efficacy," "embodied consistency," and "creativity," so that in subsequent work this architecture can undergo empirical testing and horizontal comparison with existing paradigms (such as Transformer architectures, JEPA, and active inference).

9.4 Conclusion

This paper proposes a cognitive modeling framework based on nonlinear dynamics: using a three-layer structure (Cognitive Field, Particle Flow, and Microscopic Modulation units) to characterize the basic dynamical mechanism of cognitive processes; using a slow-variable structure (GCU) to characterize the source of long-term preference and stability; constructing a decision mechanism centered on structural cost through the Benefit-Taxis Principle; using the Force Field and Benefit-Landscape to characterize the mapping from abstract biases to concrete behavior; and further using the Acceleration Anchor and Internal State filtering mechanism to accomplish the conceptual integration of the language system and the action system within a single subject. Among these, the Cognitive Field is corresponded to the spatial shape presented by the overall representation of large language models, the Association Layer is taken as the generative mechanism of the Directional Beacon, and the Internal State Curve bears the higher-level association of "whole-region activation.

At the structural level, this architecture attempts to bring thought, Internal State, behavioral selection, and long-term preference into the same dynamical picture, and to establish, through the Force Field and Benefit-Landscape structures, a mapping from abstract preference to concrete behavior; to achieve cross-modal integration of information through the acceleration curve; and to characterize, through the bidirectional response of the Association Layer, the bidirectional processes of "external stimulus → linguistic externalization" and "linguistic input → concretization of the Internal State." The resulting overall picture—physical layer corresponds to the senses, language–semantic layer corresponds to cognition, GCU kernel corresponds to the self and the Internal State, neural-network Association Layer corresponds to the mechanism of consolidating experiential connections—corresponds intuitively to the logical process of a complete living activity.

It must once again be emphasized: this architecture at present still belongs to conceptual theoretical exploration; its value resides more in conceptual unity and mechanism-abstraction capacity than in engineering-deployable system design. This paper contains no rigorous mathematical derivation, and makes no claim of any quantitative conclusion; substantial work in mathematical formalization, computational implementation, and experimental validation is still required in the future, in order to gradually test the scope and explanatory capacity of this kind of line of thought.

The standpoint of this research is not an attempt to propose a new technical pathway that can directly surpass existing methods, but—alongside the mainstream paradigm centered on data-driven approaches—to preserve and develop a line of thinking that takes the "overall operating logic of biological intelligence" as its point of departure, thereby providing a structural and cognition-rooted supplementary perspective for the long-term exploration of strong artificial intelligence. If the architecture outlined in this paper is able to provoke even a small portion of valuable further inquiry or critique in the reader's mind, its purpose will have been achieved.

Future work will study the mathematical formulation of the GCU spatial weighting mechanism, as well as the possibility of combining the hollow sphere with LLMs.

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